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Institute of Management and Career Courses (IMCC), Pune
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Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practice Assignment1

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Ass1: Create a Singly Linear Linked List with following operations

- 1. Create Linked List**
- 2. Traverse and print the node values**
- 3. Insert node at a specific position**
- 4. Find Middle node and print its value**
- 5. Delete node**
- 6. Reverse list**
- 7. Calculate the sum of every two consecutive node values.**

class Node:

```
def __init__(self, val):  
    self.data = val  
    self.next = None
```

class LinkedList:

```
def __init__(self):  
    self.head = None
```

1. Create / Append

```
def append(self, new_node):  
    if self.head is None:  
        self.head = new_node  
    else:  
        temp = self.head  
        while temp.next:  
            temp = temp.next  
        temp.next = new_node
```

2. Traverse and print

```
def print(self):  
    temp = self.head  
    count = 0  
    total_sum = 0
```



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```
nodes = []
while temp:
    nodes.append(str(temp.data))
    count += 1
    total_sum += temp.data
    temp = temp.next
print("-> ".join(nodes) if nodes else "List is empty")
print(f"Total number of nodes: {count}")
print(f"Sum of node values: {total_sum}")
```

3. Insert node at a specific position

```
def insert(self, new_node, pos):
```

```
    if pos < 1:
        print("Invalid position!")
        return
    if pos == 1:
        new_node.next = self.head
        self.head = new_node
        return
```

```
    temp = self.head
```

```
    p = 1
```

```
    while temp and p < pos - 1:
```

```
        temp = temp.next
```

```
        p += 1
```

```
    if temp is None:
```

```
        print("Position out of bounds!")
```

```
        return
```

```
    new_node.next = temp.next
```

```
    temp.next = new_node
```

4. Find middle node and print its value

```
def find_middle(self):
```

```
    if not self.head:
```

```
        print("List is empty")
```

```
        return None
```

```
    slow = self.head
```

```
    fast = self.head
```



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```
while fast and fast.next:
```

```
    slow = slow.next
```

```
    fast = fast.next.next
```

```
print(f'Middle node value: {slow.data}')
```

```
return slow.data
```

```
# 5. Delete node by value
```

```
def delete(self, val):
```

```
    if not self.head:
```

```
        print("List is empty")
```

```
        return
```

```
# If head holds the value
```

```
if self.head.data == val:
```

```
    self.head = self.head.next
```

```
    return
```

```
temp = self.head
```

```
while temp.next and temp.next.data != val:
```

```
    temp = temp.next
```

```
if temp.next is None:
```

```
    print(f'Node with value {val} not found')
```

```
    return
```

```
temp.next = temp.next.next
```

```
# 6. Reverse list in-place
```

```
def reverse(self):
```

```
    prev = None
```

```
    curr = self.head
```

```
    while curr:
```

```
        next_node = curr.next
```

```
        curr.next = prev
```

```
        prev = curr
```

```
        curr = next_node
```

```
    self.head = prev
```

```
# 7. Calculate sum of every two consecutive node values
```

```
def sum_consecutive_pairs(self):
```



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```
if not self.head or not self.head.next:
    print("Not enough nodes to form consecutive pairs")
    return []
```

```
temp = self.head
pair_sums = []
while temp.next:
    pair_sums.append(temp.data + temp.next.data)
    temp = temp.next
```

```
print("Sum of consecutive pairs:", pair_sums)
return pair_sums
```

```
def solution():
```

```
    ll = LinkedList()
```

```
    # 1. Create list
```

```
    ll.append(Node(10))
```

```
    ll.append(Node(20))
```

```
    ll.append(Node(30))
```

```
    ll.append(Node(55))
```

```
    ll.append(Node(48))
```

```
    # 2. Traverse and print
```

```
    print("--- Initial List ---")
```

```
    ll.print()
```

```
    # 3. Insert at position
```

```
    print("\n--- Insert 100 at pos 1 ---")
```

```
    ll.insert(Node(100), 1)
```

```
    ll.print()
```

```
    # 4. Find middle node
```

```
    print("\n--- Middle Node ---")
```

```
    ll.find_middle()
```

```
    # 5. Delete node
```

```
    print("\n--- Delete Node 30 ---")
```

```
    ll.delete(30)
```

```
    ll.print()
```



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6. Reverse list

```
print("\n--- Reverse List ---")
ll.reverse()
ll.print()
```

7. Sum of consecutive nodes

```
print("\n--- Consecutive Pair Sums ---")
ll.sum_consecutive_pairs()
```

```
if __name__ == "__main__":
    solution()
```

```
college.sql StringList-7-10-26.py PracticeA-9-10-26.py SinglyLinearLinkedList-3.py Ass-1SLL.py SinglyLinearLinkedList-4.py
Assignment > DSA > Ass-1SLL.py > LinkedList
18 class LinkedList:
48     def insert(self, new_node, pos):
62
63         if temp is None:
64             print("Position out of bounds!")
65             return
...
66
67     def print(self):
68         temp = self.head
69         while temp:
70             print(temp.data, end=" ")
71             temp = temp.next
72         print()
73
74     def reverse(self):
75         prev = None
76         curr = self.head
77         while curr:
78             next = curr.next
79             curr.next = prev
80             prev = curr
81             curr = next
82         self.head = prev
83
84     def sum_consecutive_pairs(self):
85         temp = self.head
86         sum = 0
87         while temp and temp.next:
88             sum += temp.data + temp.next.data
89             temp = temp.next.next
90         print(sum)
91
92     def __str__(self):
93         temp = self.head
94         while temp:
95             temp = temp.next
96         return str(self.head.data)
97
98     def __len__(self):
99         temp = self.head
100        count = 0
101        while temp:
102            count += 1
103            temp = temp.next
104        return count
105
106     def __repr__(self):
107         return str(self) + '\nLength: ' + str(len(self))
108
109     def __getitem__(self, index):
110         temp = self.head
111         while temp:
112             if index == 0:
113                 return temp.data
114             temp = temp.next
115             index -= 1
116         return None
117
118     def __setitem__(self, index, value):
119         temp = self.head
120         while temp:
121             if index == 0:
122                 temp.data = value
123             temp = temp.next
124             index -= 1
125         return None
126
127     def __delitem__(self, index):
128         temp = self.head
129         while temp:
130             if index == 0:
131                 self.head = temp.next
132             temp = temp.next
133             index -= 1
134         return None
135
136     def __iter__(self):
137         temp = self.head
138         while temp:
139             yield temp.data
140             temp = temp.next
141
142     def __contains__(self, value):
143         temp = self.head
144         while temp:
145             if temp.data == value:
146                 return True
147             temp = temp.next
148         return False
149
150     def __eq__(self, other):
151         if not isinstance(other, LinkedList):
152             return False
153         temp1 = self.head
154         temp2 = other.head
155         while temp1 and temp2:
156             if temp1.data != temp2.data:
157                 return False
158             temp1 = temp1.next
159             temp2 = temp2.next
160         return True
161
162     def __neq__(self, other):
163         return not self.__eq__(other)
164
165     def __lt__(self, other):
166         return self.__eq__(other)
167
168     def __gt__(self, other):
169         return self.__eq__(other)
170
171     def __le__(self, other):
172         return self.__eq__(other)
173
174     def __ge__(self, other):
175         return self.__eq__(other)
176
177     def __hash__(self):
178         return hash(self.__str__())
179
180     def __bool__(self):
181         return bool(self.head)
182
183     def __len__(self):
184         return len(self)
185
186     def __str__(self):
187         return str(self)
188
189     def __repr__(self):
190         return repr(self)
191
192     def __getitem__(self, index):
193         return self.__getitem__(index)
194
195     def __setitem__(self, index, value):
196         return self.__setitem__(index, value)
197
198     def __delitem__(self, index):
199         return self.__delitem__(index)
200
201     def __iter__(self):
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207     def __eq__(self, other):
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